

KayLynn Larrison

Curriculum Vitae

RESEARCH INTERESTS

Community Resilience | Climate Adaptation | Spatial and Quantitative Methods | Environmental Justice

EDUCATION

Ph.D. Sociology, Louisiana State University 2022 - 2026

Dissertation: *Climate Change Attributable Flooding and Post-Disaster Housing Market Outcomes: Evidence from Hurricane Harvey*

M.S., Sociology, University of North Texas 2020 - 2022

B.S., Sociology, Texas A&M University 2017 - 2020

PROFESSIONAL EXPERIENCE

2025 – present Resilience Fellow (co-hosted by the Community Resilience Center at The Water Institute and the Southern Mississippi Planning and Development District), SBP, Gulfport, MS

- Assists in developing state, regional, and federal grant applications for infrastructure and resilience-related projects in coastal Mississippi; more than \$6 million in external funding has been secured to date.
- Conducts vulnerability assessments, collaborates with local officials, and presents at public meetings to inform local resilience and Community Rating System (CRS) planning efforts.
- Integrates and analyzes NFIP claims and policies, flood hazards, wetlands, land use, shoreline conditions, infrastructure, and demographic data using R and ArcGIS.
- Assists with planning community events and creating outreach materials focused on flood risk and hazard preparedness.

2024 – 2025 Graduate Research Assistant (Dr. Kevin Smiley), Louisiana State University, Baton Rouge, LA

- Collaborated with researchers from LSU, NCAR, and The Water Institute on interdisciplinary climate attribution research.
- Developed and validated a hierarchical geospatial dataset linking modeled wind and storm-surge damages with demographic characteristics for nearly one million structures in southern Louisiana.
- Conducted statistical and spatial analyses examining climate change attributable hurricane impacts and socioeconomic inequality.
- Produced reproducible analyses, maps, figures, and research products for peer-reviewed publication.

- Designed and delivered an applied R mini-course covering data management, analysis, and visualization.
- 2022 – 2023 Graduate Assistant (Dr. Rhiannon Kroeger/Dr. Kevin Smiley/Dr. Tim Slack), Louisiana State University, Baton Rouge, LA
- 2020 – 2022 Technical Writer, Pacston Technologies
- 2019 – 2020 Research Assistant, Stuart J. Hysom Social Psychology Laboratory (Dr. Jane Sell/Dr. Katie Constantin), Texas A&M University, College Station, TX

METHODS & SKILLS

GIS & Spatial Analysis: R (sf, terra), ArcGIS Pro, QGIS, spatial data integration, raster and vector analysis, geoprocessing, georeferencing, and cartography.

Quantitative Methods: R, regression modeling, spatial regression, multilevel/hierarchical data, survey data analysis, data management and reproducible workflows

PUBLICATIONS

Peer-Reviewed Publications

- Smiley, Kevin T., Ovel Díaz-García, James M. Done, Scott A. Hemmerling, KayLynn Larrison, Rubayet Bin Mostafiz, Md Adilur Rahim, Alexandra N. Ramos-Valle, and Jonathan Teye Yevuyibor. (2026). “Attributing multi-hazard impacts of Hurricane Ida in south Louisiana to past, present, and future climates: Advancing extreme event impact attribution.” *Environmental Research Letters*. <https://doi.org/10.1088/2752-5295/ae6889>.
- Smiley, Kevin T., Illan Noy, Michael F. Wehner, Christopher C. Sampson, Oliver E.J. Wing, and KayLynn Larrison. (2025). “Climate Change and Federal Aid Disbursements After Hurricane Harvey: An Extreme Event Attribution Analysis.” *Environmental Research Letters*. <https://doi.org/10.1088/1748-9326/adf978>.
- O’Connell, Heather, Marcus Venable, and KayLynn Larrison. (2025). “Neighborhood Disadvantage and the Geographic Distribution of Registered Sex Offender Concentration: Another Dimension of Racial Inequality?” *Race and Social Problems*. <https://doi.org/10.1007/s12552-025-09446-0>.

Extension Products

- Collini, R., B. Mehney, and K. Larrison. (2026). “Flood Event Probability”. Community Resilience Center at The Water Institute.
- Haertling, A., K. Larrison, B. Mehney, M. Daigle, N. Pace, and D. Reid. (2025). “Program for Public Information.” Community Resilience Center at The Water Institute.

SUCCESSFUL FUNDING

EPA Brownfield Coalition Grant

\$1.5 million (Data analysis & grant writing)

Revitalizing Historic Stone County

National Heritage Area Community Grant

\$10,000 (Grant writer)

Stone County Airport Fixed-Based Operator

State Economic and Infrastructure Grant

\$250,000 (Grant writing)

Stone County Industrial Park Sewer Improvements

Gulf Coast Restoration Funds

\$3.2 million (Grant writing & evidence development)

George County Industrial Park Water Improvements

Gulf Coast Restoration Funds

\$1.3 million (Grant writing & evidence development)

Perry County Sportsman's Lake Wharf & Floating Dock

Mississippi Outdoor Stewardship and Trust Fund

\$32,610 (Grant writing & implementation planning)

Pending Proposals

Jackson County Stormwater Drainage Improvements

FEMA Building Resilience Infrastructure and Communities

\$9.4 million requested (Developed and implemented community engagement and data analysis strategy)

Harrison County Firetruck Apparatus Replacement

FEMA Assistance to Firefighters

\$2.2 million requested (Grant writer)

TEACHING EXPERIENCE

Summer 2025	Designed and delivered a summer mini course focused on data management and visualization in R.
Fall 2023	Supplemental Statistics Instructor (Sam Ramey), Louisiana State University, Baton Rouge, LA
2022	Teaching Assistant, Medical Sociology (Dr. Rhiannon Kroeger), Louisiana State University, Baton Rouge, LA
2021 - 2022	Teaching Assistant (Dr. Karen Gregg/Dr. Deborah Fessenden), University of North Texas, Denton, TX

CONFERENCE PRESENTATIONS

- Bohrson, Michelle, Sherry Risk, Sara Hambrick, and KayLynn Larrison. (2025). "An Alternative Model for Enhancing Community Resilience." Natural Hazards Workshop Practitioner's Meeting. Broomfield, Colorado.
- Larrison, KayLynn and Kevin Smiley. (2025). "Effects of Climate Change Attributed Damage and Flooding on Housing Stability." Natural Hazards Workshop Researcher's Meeting. Broomfield, Colorado.
- Smiley, Kevin T., Teye Yevuyibor, KayLynn Larrison, James Done, and Alexandra N. Ramos-Valle. (2025). "Social Implications of an Extreme Event Attribution Analysis of Hurricane Ida's Wind and Flood Impacts." American Meteorological Society. New Orleans, Louisiana.
- Smiley, Kevin T., James Done, Teye Yevuyibor, Alexandra Ramos, and KayLynn Larrison. (2024). "Investigating Socio-spatial Impacts from an Extreme Event Attribution Analysis of Hurricane Ida." American Geophysical Union. Washington DC.
- Smiley, Kevin T., Teye Yevuyibor, KayLynn Larrison, James Done, and Alexandra N. Ramos-Valle. (2024). "Social Responses to Climate Change Attributed Flooding in South Louisiana." Natural Hazards Workshop Researchers Meeting. Broomfield, Colorado.
- Larrison, KayLynn. (2024). "Threat perceptions and group empathy: An explanation of intergroup attitudes." Southern Sociological Society annual meeting. New Orleans, Louisiana.

SERVICE ACTIVITIES

2026	Gulf Resilience Community of Practice Science Café Co-lead
2026	Reviewer <i>Cities</i>
2025	LSU Fred C. Frey Memorial Award Committee Member
2025	LSU Discover Day Undergraduate Research and Creativity Symposium Judge
2024	Reviewer <i>GeoHealth</i>
2024	Southern Sociological Society Local Arrangements Committee Member
2024	LSU Discover Day Undergraduate Research and Creativity Symposium Judge

AWARDS

2025	Roland Pellegrin Outstanding Graduate Student in Sociology
2025	Environmental Data Science Innovation & Impact Lab (ESIIL) Innovation Summit, University of Colorado, Travel Grant
2025	Compounding, Cascading, and Critical Risks to U.S. Infrastructure and Security Workshop, Portland State University, Travel Grant
2024	CyberTraining for Disaster Management Workshop Travel Grant